Supplementary Material for Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

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A. The allocation-channel study, in brief

This appendix reports, with their numbers, the three findings of our allocation-channel study — controlled resampling-prior experiments on the same pipeline, scenes, and budgets, conducted before moving topology into the objective — that this paper builds on.

(1) Geometric metrics are topology-blind. Three constructed cases each pit a topologically correct candidate A against a wrong one B whose geometry is bisection-matched, making Chamfer equal by construction (40k samples, metrics normalized by the ground-truth bounding diagonal; Table S1). The discriminating-dimension bottleneck separates every pair — by 35–40× in the H0/H2 cases — while in the first two cases Hausdorff$_{95}$ actually prefers the topologically wrong candidate. Chamfer cannot tell a benign bump from a destroyed void; the persistence metric can.

(2) The prior arms, and why voids are "width-primary". The study biases DiffSoup's own resampler with a precomputed importance field built from the target's significant features. Each significant feature f (lifetime p_f above the significance threshold) contributes Gaussian kernels at the coordinates C_f of its birth/death-simplex vertices — the loop-filling triangle, void-filling tetrahedron, or component-merging edge — sized by the feature's own death scale:

$$\phi_d(q) = \sum_{f:\,\dim f = d} \sum_{c \in C_f} w_f \, e^{-\|q-c\|^2/(2\sigma_f^2)}, \qquad w_f = p_f/s^2,$$
$$\sigma_f = s\,\mathrm{clip}\big(\sqrt{\mathrm{death}_f},\, 3r_{\mathrm{med}},\, \tfrac{1}{2}\mathrm{diag}\big), \tag{1}$$

evaluated on a dense target-surface sample, normalized per dimension to $[0,1]$ (99.5th percentile), combined by max; a query takes its nearest surface sample's value. The spread knob s widens each kernel while preserving its injected surface mass ($\sigma \to s\sigma$, $w \to w/s^2$, so $w\sigma^2$ is s-invariant): $s{=}1$ is the concentrated prior (B2), $s{=}3$ the spread prior (B4) — the field C5 reuses. The field enters resampling at strict budget parity through two levers: prune protection — faces drop in order of vis + $\lambda\, Q_{0.9}(\mathrm{vis}) \cdot \mathrm{imp}$, removing exactly the required count — and respawn bias — top-importance visible faces subdivide first, then the protective order re-prunes to the cap. Remaining arms: B0 baseline; B1 = B2 applied at initialization only; B3/B5 random-center non-topological controls width-matched to B2/B4 respectively. On the void class (Table S2; $N{=}1200$, five paired seeds) B4 cuts the bottleneck 33–53% below baseline — but the width-matched random control B5 recovers most of that gain, leaving only a small, shape-dependent topological residual (cylinder 4.6σ, sphere 2.2σ, cube tie); one-shot placement washes out entirely (B1 ≈ B0, .0546 vs .0535 on the sphere). Hence the verdict quoted in the main paper's introduction: the prior's topological value is carried largely by its spatial width.

(3) No prior shape repairs loops. On the torus ($N{=}700$, five paired seeds, true $\beta_1{=}2$) every arm fails the topology-specificity test: the best condition is the non-topological wide control B5 (.0267), ahead of spread-topological B4 (.0316), baseline (.0409), and narrow random B3 (.0397); the concentrated topological field B2 is worst (.0425) and manufactures phantom handles — final significant-H1 count 4.4 against the true 2.0, with the worst Chamfer (1.32 vs baseline 1.01). Spreading merely stops the harm; it never beats the random control. That failure is the opening premise of the main paper, and the class the loss channel wins there (main paper §5.1, Table 1).

Table S1. Blindness probes: A topologically correct, B wrong, geometry bisection-matched so Chamfer is equal by construction. Bottleneck to ground truth in the discriminating dimension.

case (disc. dim)	Cham.% A = B	bott. A	bott. B	ratio
thin handle, genus 0→1 (H1)	0.916	≈0	.0302	—
bridge merges two components (H0)	0.628	.0014	.0574	40×
punctured enclosed void (H2)	0.977	.0040	.1385	35×

Table S2. Allocation study, void class (H2), N=1200, five paired seeds: tail bottleneck to target (lower is better). Spread beats concentrated; the width-matched random control (B5) recovers most of the spread gain.

shape	B0	B2 conc.	B3 narrow rand.	B4 spread	B5 wide rand.
sphere	.0535	.0438	.0440	.0357	.0374
cube	.0579	.0409	.0403	.0273	.0270
cylinder	.0545	.0440	.0477	.0360	.0402

Table S3. Significant bars (H1 / H2) on the clean target surfaces vs. sampling density M; floor = $6\,r_{\mathrm{med}}(M)$, seed 0.

	M=512	M=1024	M=2048	M=4096
sphere	0 / 1	0 / 1	0 / 1	0 / 1
torus	1 / 0	1 / 0	2 / 0	2 / 1
double torus	0 / 0	2 / 0	2 / 0	4 / 0

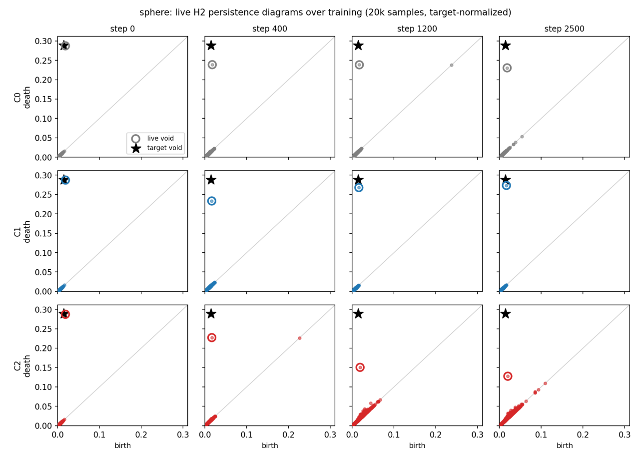

Figure S1. Live H2 persistence diagrams over training (one sphere seed; 20k samples, target-normalized; star = target void bar, circle = most persistent live bar). Top, baseline: the void drifts off target. Middle, loss: the void returns to the target once the ramp activates. Bottom, repulsion control: the void is pinned and dragged away while near-diagonal phantom bars accumulate.

B. Diagnostics: density sensitivity and diagram trajectories

Sampling density vs. measurability. Table S3 counts the significant bars of the clean target surfaces across sampling densities under the self-calibrating floor $6\,r_{\mathrm{med}}(M)$ of the main paper's §6. It confirms the three density claims made in the paper: the sphere's void clears the floor at every tested density; the torus needs M=2048 before its second loop is measurable, and its own void becomes measurable only at M=4096 (hence the H1 restriction at M=2048); and the double torus's tube loops cross the floor between M=2048 and M=4096 — consistent with the $M \gtrsim 3{\times}10^3$ estimate there — so all four loops are measurable at M=4096.

Diagram trajectories: the pinning mechanism, visualized. Fig. S1 shows the live H2 diagrams of one sphere seed under C0/C1/C2 at four training steps. All three arms start with the void essentially on target (initialization samples the SfM point cloud), and photometric training alone drags it away (the baseline settles ≈.06 below). The loss arm dips identically while the ramp is inactive, then pulls the void back to within ≈.016 of target — matching its measured tail. The control arm is qualitatively different: the void's death is dragged progressively down (≈.23 → .15 → .13) while a carpet of phantom mid-persistence bars accumulates near the diagonal — the repulsion-equilibrium signature discussed in the main paper's §6.

C. Additional result figures and the repair gate

Fig. S2 shows the generality-wave training trajectories (main paper §5.2, Table 2); Fig. S3 the per-seed tail spread behind the sphere row of main-paper Table 1.

The stage-3a gate (main paper §5). Fig. S4 shows three of the four repair probes run before integration: the loss alone (no photometric term, Adam directly on point positions) must repair defective clouds. The predicted location-blind pathology of recruitment did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned),

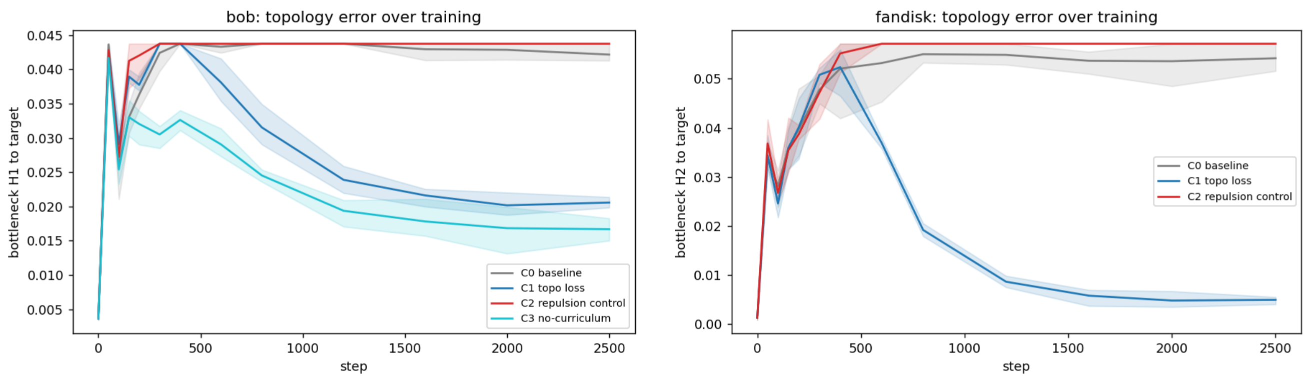

Figure S2. Generality trajectories (line = seed mean, band = seed min–max). Left: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with #sig H1 = 2 in every seed, while the control rides the baseline after collapsing a loop. Right: fandisk (CAD) — the largest effect measured in this study (10.4×).

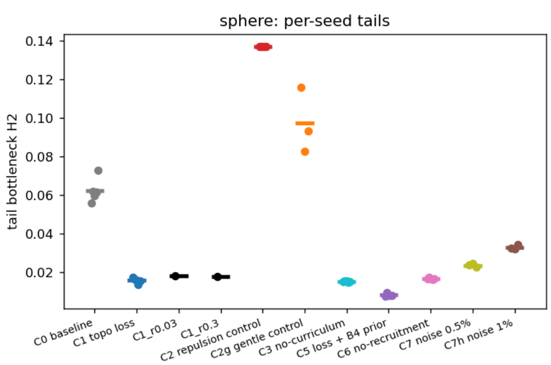

Figure S3. Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from above; the ρ ablation arms are indistinguishable from ρ=0.1.

while cutting the 1-D bridge raises the recruited bar's death monotonically toward target — the runaway concentrates on the only profitable direction.

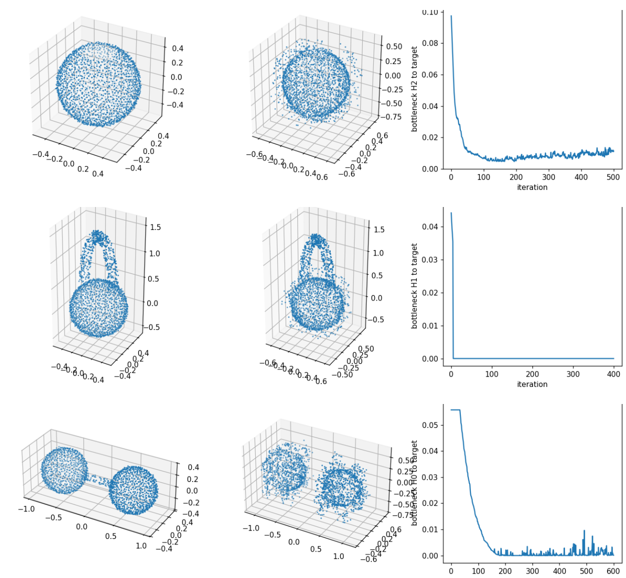

Figure S4. The stage-3a gate: the loss alone (Adam on raw points, no photometric term) repairs defective clouds; each row shows initial cloud, final cloud, and bottleneck-to-target trajectory. Top (T1): punctured sphere — the matched birth term closes the void (8.8×). Middle (T2): spurious handle — the diagonal term kills the extra H1 bar. Bottom (T4): bridged merge with no significant live H0 bar — recruitment severs the bridge (450×, β_0 1→2), the case where plain optimal matching provably has zero gradient. (T3, mis-spaced components, 1229×: omitted for space.)

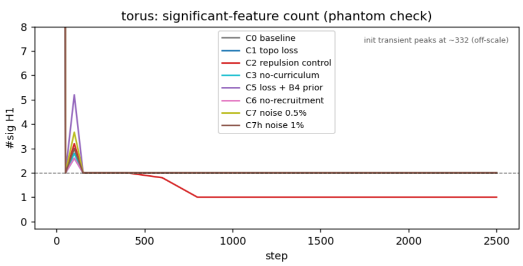

Figure S5. Torus phantom check (main paper §5.1): significant-H1 count over training (seed means; dashed line: true β_1=2). After the brief resampling transient near step 100, every loss arm sits exactly on two significant loops — zero phantom handles — while the repulsion control (C2) collapses a loop by step ∼800 and never recovers it.